

# Literature Review

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## 1 Terminology

**Morph/Morphing** - When I use the word morph here I am referring to the process of polygons, curves, or surfaces transforming through some interpolatory process.

**Homotopy** - two continuous functions from one topological space to another are called homotopic if one can be "continuously deformed" into the other, such a deformation being called a homotopy.

**Cubic B-Splines** - A parametrically defined curve represented by a series of control points, which although not interpolated, control the shape and tangents of the curve. Each segment is controlled by 4 points of the control polygon. [Li21]

**NURBS** - Non-Uniform, Rational B-Splines.

**PSLG** - Planar straight line graph. Contained within a 2D plane, consisting of non-intersecting except possibly at endpoints, lines and points.

**Knot Vector** - A sequence of parameter values that define the positions of knots in a mathematical curve, used here on B-Splines and more generally, NURBS.

**Mathematical Knot** - A closed curve in some space with a number of twists that topologically cannot be undone without cutting the curve.

**Thin-Plate Splines** - Splines formulated with certain smoothness restrictions so that the surface remains stiff but smooth to a certain degree.

**Fréchet Distance** - A measure of similarity between two curves by connecting said curves by straight ropes of sorts and summing the distances.

**Blend Shapes** - A method of 3D computer animation used together with techniques such as skeletal animation. In a morph target animation, a "deformed" version of a mesh is stored as a series of vertex positions. In each key frame of an animation, the vertices are then interpolated between these stored positions.

## 2 Main Idea – Inspiration & Motivation

Engineers, artists, and especially mathematicians would love to have a method for which to morph between 2 defined curves where the interpolatory curves remain at least  $C^1$  continuous. Back in the 80's and 90's these methodologies were somewhat popular in the field of graphics and geometric modeling in the form of morphs between various surfaces that describe the human face [Yue+15]. What now seems a little archaic, morphing found an early, almost natural, use case when animating the human face. Artists would like to interpolate between some key frames and have some program figure out what the surface should look like in between. Some approaches that popped up when dealing with mesh faces includes the ever-popular blend shapes method where you just linearly interpolate between some series of meshes. This method is so good in fact that it is still seen use today. But some of the other methods that were experimented with was a method to interpolate B-Spline surfaces in an attempt to generalize the morph beyond the need for facial morphs and across any B-Spline surface.

Getting to the main idea now, what we really want to do is take two shapes, generate some intermediary shapes as to provide a "nice" transition between them. A paper by Dym has a really nice approach to the problem from a topology point of view. In practice, "Morphing of curves is typically divided into two subproblems. The vertex correspondence problem ... [and] the vertex path problem" [DSL15]. So basically we need to do 2 main things, first, find a mapping of points from one curve to the other. This may involve adding or removing vertices to our control polygon as we interpolate to achieve the best results. And second, actually finding the mapping of intermediary curves that "nicely" interpolate the given curves.

Some basic requirements that usually constitute a "nice" morph is that people typically disallow any cusps and sharp edges in the intermittent frames. Even stronger so, a lot of people tend to want at least  $C^1$  continuity on all interpolatory curves. There even exists methods to go beyond this and restrict the inner frames to only  $C^2$  curves [BC16]. The paper, BC16, is interesting as it describes a method to morph  $C^1$  cubic splines into  $C^2$

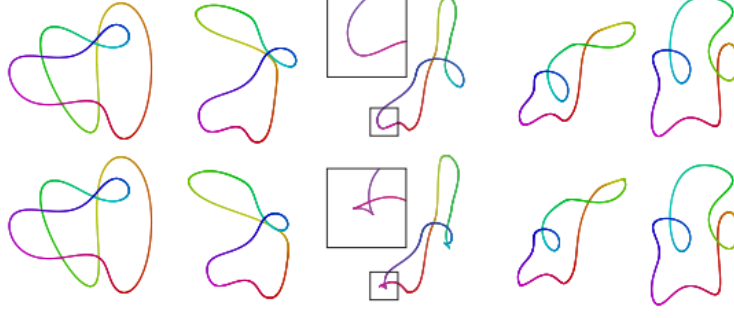


Figure 1: [DSL15] 2 different morphs between curves, one with and one without a cusp in an intermediary curve.

curves. It provides a homotopic deformation for any at least  $C^1$  cubic B-Spline into a spline at  $C^2$  continuity by reordering the know vector while minimizes large changes in the definition of the curve.

### 3 The Math

For specifically the case of rational B-splines, Tao Ju and Ron Goldman have a pretty great method of using the weight vectors associated with the B-splines to dictate their morph [JG03]. They first start with a linear interpolation of two B-spline's control points and weights of those control points. This method is ok, but not ideal as the results it produces may lead to self-intersections or "shrinking", where pieces of the curve bunch up as the weights between the control points do not line up correctly for where they are positioned and appear to bunch up the curve. The intuition here is to consider redefining operations on the B-Spline in Grassmann space, where each control point has an associated mass and operation like addition and multiplication are redefined in order to model realistic behavior between the masses. Here the paper defines a new notation to handling mass computation, called mass-point form:  $\frac{mP}{m}$  where  $m$  is the mass and  $P$  is the point in the control polygon of the B-spline. The paper then redefines a couple operations:

$$\begin{aligned} \text{Scalar Multiplication: } c \otimes \frac{mP}{m} &= \frac{cmP}{cm} \\ \text{Addition: } \frac{m_1P_1}{m_1} \oplus \frac{m_2P_2}{m_2} &= \frac{m_1P_1 + m_2P_2}{m_1 + m_2} \end{aligned}$$

In practice, scalar multiplication of the weights now does not update the positions of the points and addition is similar to summing the masses and moving the new point to the center of mass between the two. At this point the paper now reformulates the representation of a rational B-spline using the new definitions present in Grassmann space:

$$P(u) = \frac{\sum_{k=0}^p w_k P_k N_k^n(u)}{\sum_{k=0}^p w_k N_k^n(u)} \Rightarrow \sum_{k=0}^p \left( \frac{w_k P_k}{w_k} \right) \otimes N_k^n(u)$$

Where  $P_k$  is the collection of control points  $w_k$ , the weights, and  $N_k^n J(u)$  the B-spline basis functions. The trick now, is that you can now perform a linear interpolation between your starting curve and target curve in Grassmann space and project onto affine space. Points are now attracted to other points with the larger weight now, so mass should distribute in such as way as to appear more natural when interpolating between rational B-splines. This method described only works on uniform rational B-splines, but the paper also extends these formulas for the non-uniform cases for the knot-vector. Overall I believe that this paper does an amazing job at providing such a simple technique that drastically improves the outcome of interpolatory frames between NURBS and Tensor Products surfaces.

Similar in execution, yet not in function, Gomez et. al. have a wonderful paper detailing free-form deformations on CAD geometry using B-Spline transformations [GGH]. The main use in this method is to convert from parametric CAD geometry to B-reps. The way it goes about this is by applying a penalty function on the control polygon for the underlying B-Spline for a surface. The context of this paper is for designing plane wings, so it makes sense to model using B-Splines, but then needing to convert to some volume limitation model. The penalty function works by drawing boundaries through the neighbors of each control point as a kind of diagonal grid across the patches as pictured: The way the Free-Form Deformation (FFD) works is by embedding the

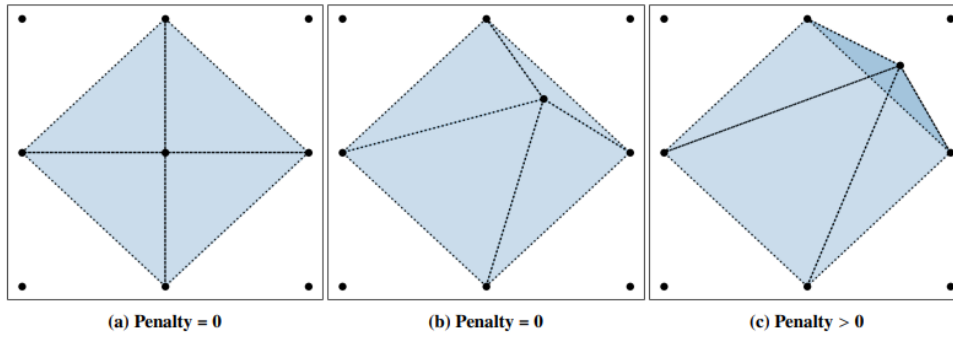


Figure 2: [GGH] Penalty based on interior FFD (Free-Form Deformation) control points

curves in a grid of points defined by a tensor product trivariate B-spline where the embedding exists in a higher space topologically than the object that is embedded. Since this case restricts self-intersections, they actually have less degrees of freedom as traditionally deforming the B-Spline representation. On the other hand they actually have a broader range of shapes that can be formed easier with the lower degrees of freedom which also help with optimizations.

One of the other processes I can take is to physically model the B-spline in some physics engine, then performing some controlled simulations to get the "morph" behavior that I desire. This can be accomplished through the details provided in the paper, "Physical Modeling with B-spline Surfaces for Interactive Design and Animation" [TC90]. This is a paper from 1990, but I still believe is very applicable to current physical modeling processes and reminds me a lot of modern work by Matthias Müller in tetrahedralized soft-body volumes and fluid simulation. In this paper though the control polygon for the B-Spline becomes a type of elastic cage that deforms the B-Spline. By using a series of "springs" between the control points and "hinges" along the edges of the control points, the authors have presented a method that apparently can stably model an elastic B-Spline. Of course there are other considerations in this model such as the damping force. The idea and application of such a method is to model a kind of semi-rigid structure like a sheet of rubber or plastic.

## 4 Related Work

When speaking of morphing more broadly, you end up looking at deformable surfaces in general. In a great review paper by Montagnat, they briefly discuss the history of the many techniques for which people deform geometric surfaces [MDA01]. The paper is a little dated as it was released in 2001, but still very relevant for as I have mentioned, the core foundational research in this field took place in the 80's and 90's. One thing the paper does really well is splitting the field into various types of geometric representations of deformable surfaces. To give a brief overview, basically, the work I am trying to accomplish in this project is focused on morphing of B-splines which is a type of "Polynomial finite support representation" of a deformable surface. This is of course an explicit representation of the surface and is a continuous model. By contrast there also exist implicit models and discrete models. Some other examples include an algebraic surface for an implicit model or triangulation and meshes in general for a discrete representation. The applications of the various methods is almost always dependent on the context of the specific problem that you want to solve. Early I had mentioned the motivation of morphing as wanting to work interpolate facial expression when animating human characters. This ended up becoming a discrete mesh deformation problem because meshes are what artists actually wanted to animate.

There is also some related work when talking about general deformable surfaces in deformable volumes that took place in the late 1980's. One interesting technique to note is a general method for defining these warping transformations on B-reps and CSG's presented "Free-Form Deformation of Solid Geometric Models" [SP86]. This paper describes a shape in some file format by the tensor product, then adds its own control points for a cage of sorts which it can use to warp the space within, including the volume which gets warps as a result. This method has some immediately applicable applications where they can use these new warps to define more interesting, curved shapes that were otherwise extremely difficult to model or store in a CSG for instance. Taking this process a step there is another paper, using the previous as a baseline to design a brand new tool to deform these objects in an effort to apply current research to a functional consumer modeling tool [Coq90].

In the field of Computational Geometry, arbitrary morphs show up a little more frequently, specifically when attempting to solve problems like providing some morph or interpolation between two given planar straight line graphs (PSLG). As an introduction to some of these topics, the paper by Buchin et. al. gives a nice presentation of ideas on morphing and how they might be used in computational geometry through their presentation on

morphing arbitrary geometries through higher dimensions to get smooth transformations [Buc+23]. The main point of interest in this showcase is that by mapping the morph from 2D, the planar graph, through 3D space, they can extend the morph to two possibly non-topologically equivalent PSLG's. This is interesting news for us for if we are working on morphing 2D closed-curve splines with different turning numbers, then it may be possible to spin it around in 3D to untie the mathematical knot.

Another more unique work lies in medical applications. In the paper Quantitative Coronary Angiography with Deformable Spline Models, the authors use some B-Spline fitting methods to deform a spline surface around the shape of medical imaging of blood vessels [KLA97]. They then use frequency information extracted from the splines to measure likelihood of Coronary artery disease where the artery thins out in some locations, restricting bloodflow and increasing likelihoods of tearing. I felt like including a more odd method like this because it represents another type of optimization problem with deformable splines with a more unique application despite this research being outdated and the techniques if still in practice have more than likely been replaced with AI systems.

Another interesting piece of related work is a methodology based on Fréchet distances. This leads to a type of locally optimal morphing pattern by pulling together points pseudo-linearly based on distance to goal points for each control point. There's an interesting paper that explores this relationship and improves on the simple linear approach by measuring distance based on travel time along the overlapping control polygons, measuring distances with the knot vectors [HRR24].

## 5 Current Research & Addition Application

A lot of the current research in this discipline is mostly centered in one of two areas, either artistic expression such as animation or simulation, trying to get better interpolatory curves or in pure mathematics fields, attempting to define methods or prove existing methods or generating a morph between arbitrary curves or surfaces. One of my inspirations for this project comes from a more artistic paper, detailing the creation of shapes and semi-polygons derived from circles and blended polyhedrons derived from spheres [Ver25]. These so called Curvygons and Curvyhedra enumerate various morphs between circular objects and polygon-like objects by first starting with a circle for instance in the 2D case, segmenting it into a piecewise, closed, NURBS, and extending each control point weight towards infinity so that the sphere approaches the shape of the circumscribed polygon. The

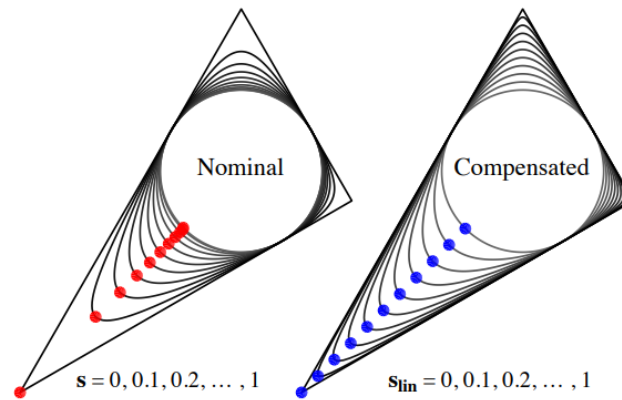


Figure 3: [Ver25] The "Curvygons" as the weights of the control polygon changes.

above graphic is also interesting, so as to support any more non-technical user, such as most artists or graphic designers who would like to use these curvygons, the paper details a compensated mapping so that the weights between zero and one of the control points correspond to the circle at zero and outer polygon at one. Another interesting thing to note, I will not get deep into the math here, but the polygon is also formulated in such a way that the weights on the control points on the NURBS are all weighted equally with a linear interpolation between the polygon and circumscribed circle.

Another interesting application of similar techniques is in "Swarm Formation Morphing for Congestion Aware Collision Avoidance" [Yas+20]. This time by using thin-plate splines the authors here construct a morph that aims to avoid obstacles to allow members of a "swarm" to pass by each other with ease, more properly simulating swarming patterns in reality. I don't really know how a method like this stacks up against some popular methods for swarming behaviors like boids, but I feel like some of the theory here could improve boid behavior at a realistic global scale.

It is also worth mentioning in current research that there are a few morphing methods that approach the problem with neural representations. I did not really look into these areas too much as it was not my initial

interest, but if time permits, then I do need to look into what work is happening here.

## References

- [SP86] Thomas W. Sederberg and Scott R. Parry. “Free-form deformation of solid geometric models”. In: *SIGGRAPH Comput. Graph.* 20.4 (Aug. 1986), pp. 151–160. ISSN: 0097-8930. DOI: 10.1145/15886.15903. URL: <https://doi.org/10.1145/15886.15903>.
- [Coq90] Sabine Coquillart. “Extended free-form deformation: a sculpturing tool for 3D geometric modeling”. In: *SIGGRAPH Comput. Graph.* 24.4 (Sept. 1990), pp. 187–196. ISSN: 0097-8930. DOI: 10.1145/97880.97900. URL: <https://doi.org/10.1145/97880.97900>.
- [TC90] Jeffrey A. Thingvold and Elaine Cohen. “Physical modeling with B-spline surfaces for interactive design and animation”. In: *SIGGRAPH Comput. Graph.* 24.2 (Feb. 1990), pp. 129–137. ISSN: 0097-8930. DOI: 10.1145/91394.91430. URL: <https://doi.org/10.1145/91394.91430>.
- [KLA97] A.K. Klein, F. Lee, and A.A. Amini. “Quantitative coronary angiography with deformable spline models”. In: *IEEE Transactions on Medical Imaging* 16.5 (1997), pp. 468–482. DOI: 10.1109/42.640737.
- [MDA01] J. Montagnat, H. Delingette, and N. Ayache. “A review of deformable surfaces: topology, geometry and deformation”. In: *Image and Vision Computing* 19.14 (2001), pp. 1023–1040. ISSN: 0262-8856. DOI: [https://doi.org/10.1016/S0262-8856\(01\)00064-6](https://doi.org/10.1016/S0262-8856(01)00064-6). URL: <https://www.sciencedirect.com/science/article/pii/S0262885601000646>.
- [JG03] Tao Ju and Ron Goldman. “Morphing Rational B-spline Curves and Surfaces Using Mass Distributions”. In: *Eurographics 2003 - Short Presentations*. Eurographics Association, 2003. DOI: /10.2312/egs.20031046.
- [DSL15] N. Dym, A. Shtengel, and Y. Lipman. “Homotopic Morphing of Planar Curves”. In: *Computer Graphics Forum* 34.5 (2015), pp. 239–251. DOI: <https://doi.org/10.1111/cgf.12712>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.12712>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/cgf.12712>.
- [Yue+15] Mei-Heng Yueh et al. *Conformal Surface Morphing with Applications on Facial Expressions*. 2015. arXiv: 1504.00097 [cs.GR]. URL: <https://arxiv.org/abs/1504.00097>.
- [BC16] Michael Bartoň and Victor Manuel Calo. “Gaussian quadrature for splines via homotopy continuation: Rules for C2 cubic splines”. In: *Journal of Computational and Applied Mathematics* 296 (2016), pp. 709–723. ISSN: 0377-0427. DOI: <https://doi.org/10.1016/j.cam.2015.09.036>. URL: <https://www.sciencedirect.com/science/article/pii/S0377042715004896>.
- [Yas+20] Jawad N. Yasin et al. *Swarm Formation Morphing for Congestion Aware Collision Avoidance*. 2020. arXiv: 2011.03883 [cs.R0]. URL: <https://arxiv.org/abs/2011.03883>.
- [Li21] Liang Li. “Application of Cubic B-spline Curve in Computer-Aided Animation Design”. In: *Computer-Aided Design Applications* 18.1 (2021), pp. 43–52. DOI: 10.14733/cadaps.2021.S1.43-52. URL: <https://doi.org/10.14733/cadaps.2021.S1.43-52>.
- [Buc+23] Kevin Buchin et al. “Morphing Planar Graph Drawings Through 3D”. In: *Computing in Geometry and Topology* (2023), Vol. 2 No. 1 (2023). DOI: 10.57717/CGT.V2I1.33. URL: <https://www.cgt-journal.org/index.php/cgt/article/view/33>.
- [HRR24] Sarel Har-Peled, Benjamin Raichel, and Eliot W. Robson. *The Fréchet Distance Unleashed: Approximating a Dog with a Frog*. 2024. arXiv: 2407.03101 [cs.CG]. URL: <https://arxiv.org/abs/2407.03101>.
- [Ver25] Bjorn Vermeersch. “Curvygons and Curvyhedra: 2D/3D Shapes that Smoothly Morph from Circles/Spheres to Circumscribed Polygons/Polyhedra”. In: (Feb. 2025). DOI: 10.36227/techrxiv.174060386.65988305/v1. URL: <http://dx.doi.org/10.36227/techrxiv.174060386.65988305/v1>.
- [GGH] Marlena Gomez, Marshall C. Galbraith, and Robert Haines. “Free-Form Deformation of Parametric CAD Geometry via B-Spline Transformations”. In: *AIAA AVIATION 2023 Forum*. DOI: 10.2514/6.2023-3601. eprint: <https://arc.aiaa.org/doi/pdf/10.2514/6.2023-3601>. URL: <https://arc.aiaa.org/doi/abs/10.2514/6.2023-3601>.